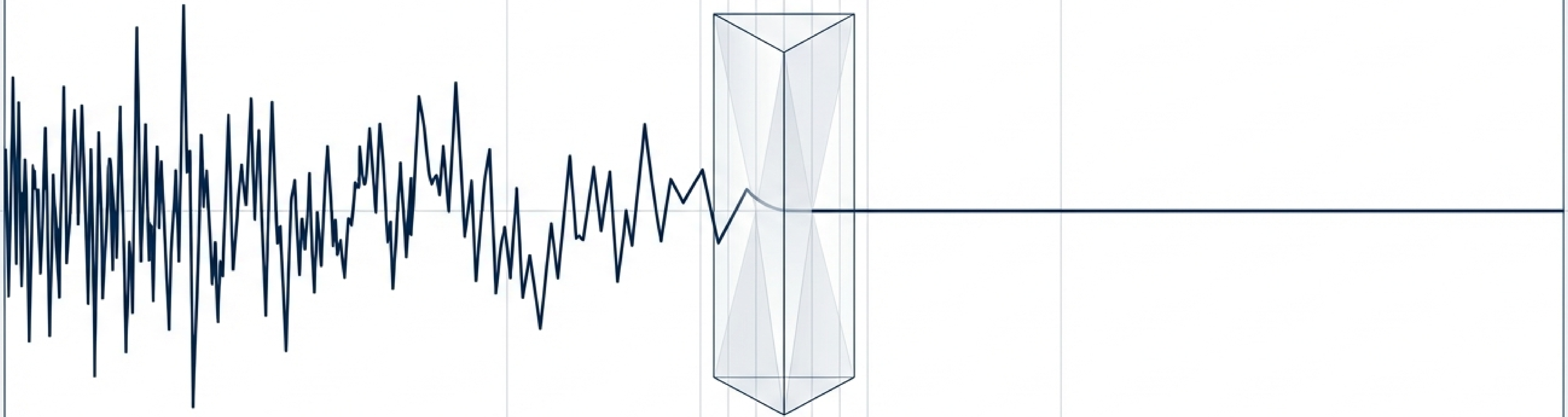
DECODING ALPHA



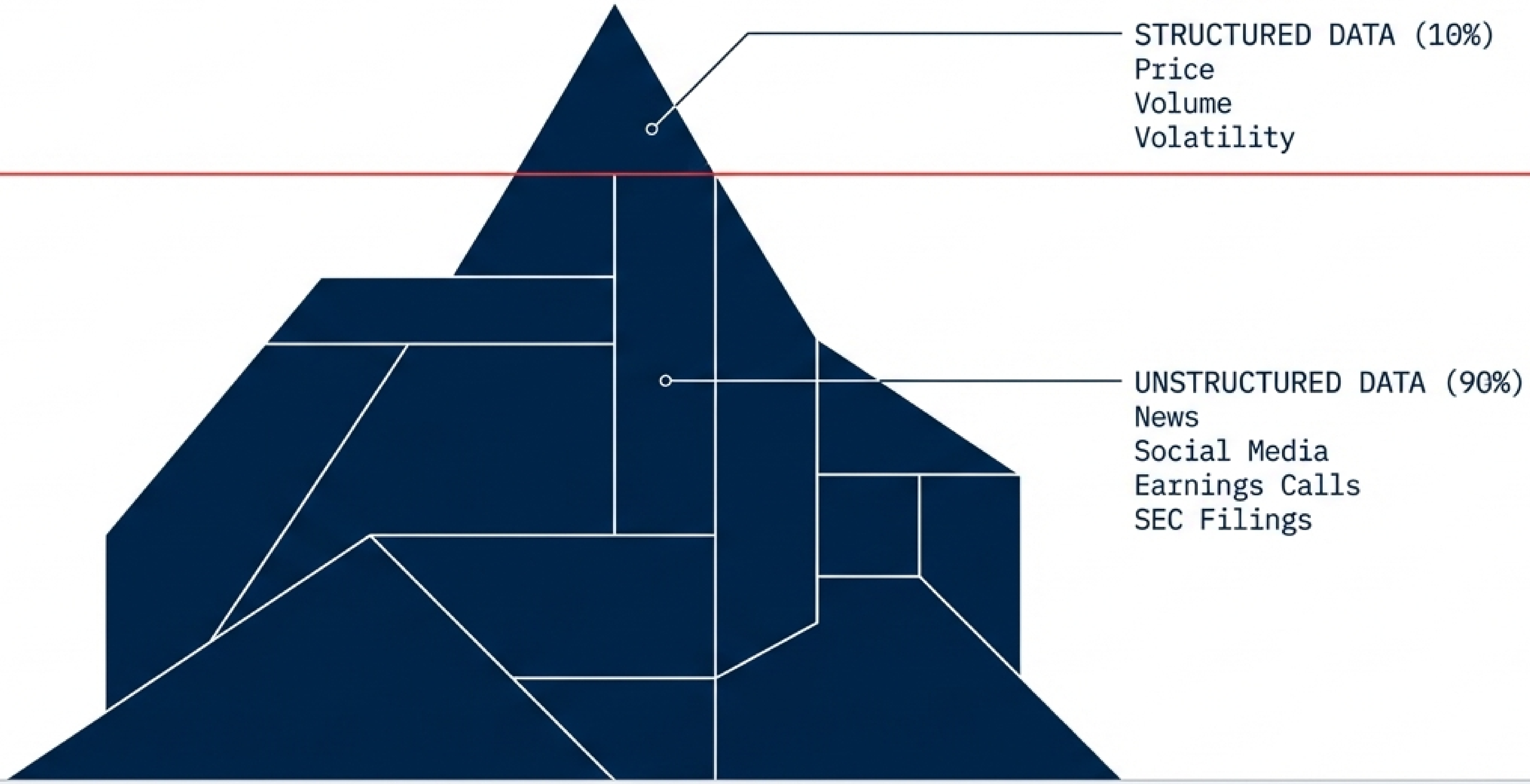
The Evolution of the Signal

From counting words to decoding the human voice: How AI is transforming financial forecasting.

Based on research by Amin et al, Roumeliotis et al, and Todd et al.

THE OCEAN OF UNSTRUCTURED DATA

Traditional quantitative models digest numbers but choke on language. The goal is bridging the gap between qualitative human expression and quantitative market movement.



THE EVOLUTION OF LISTENING

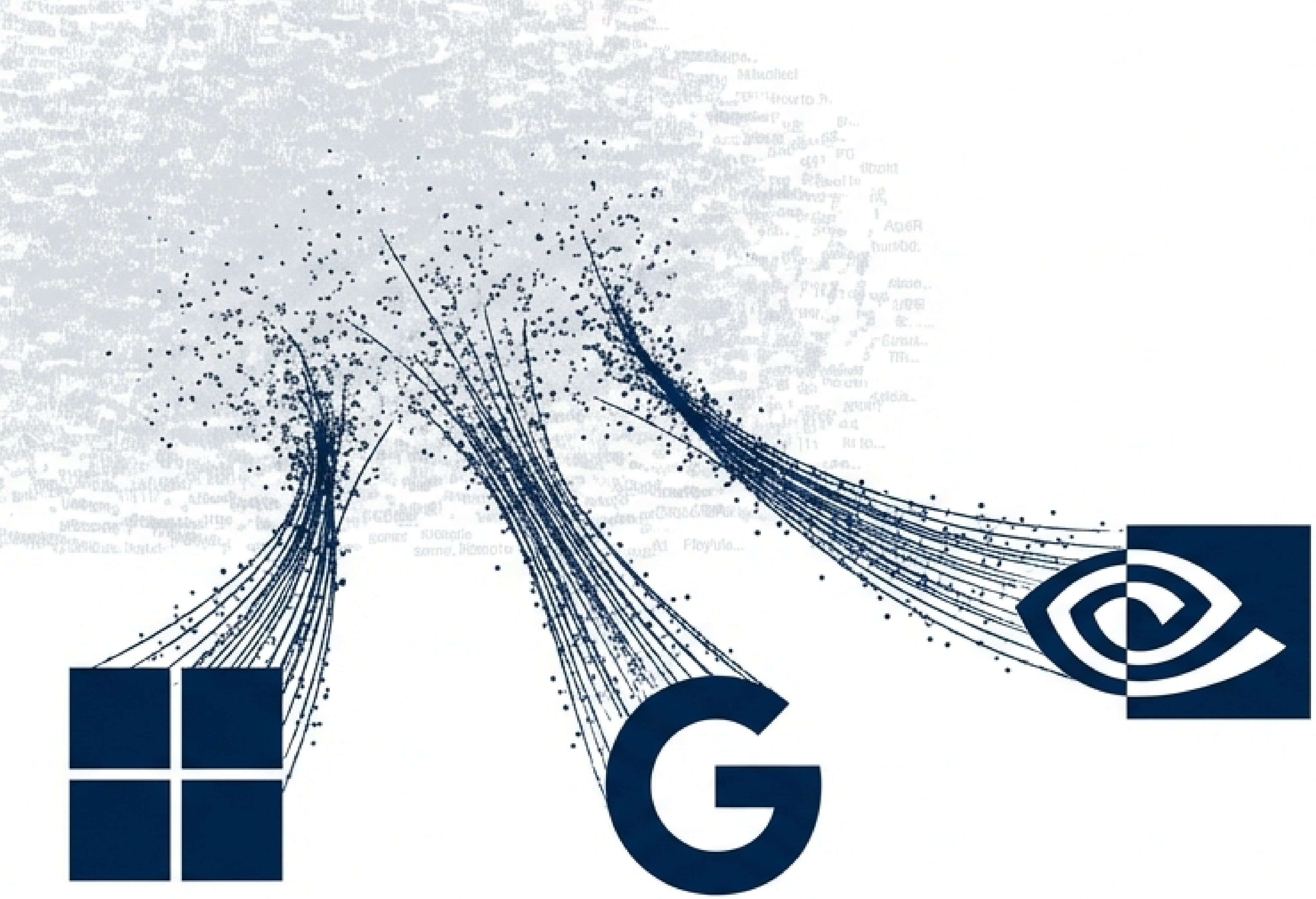
PAST
The First Filter
Dictionary Approach / Word Counting

PRESENT (A)
The Social Pulse
Machine Learning / Twitter Mining

PRESENT (B)
The Cognitive Leap
LLMs & Fine-Tuning

FUTURE
The Full Picture
Multimodal Analysis (Text + Audio)

The Hypothesis: Can Hype Predict Value?

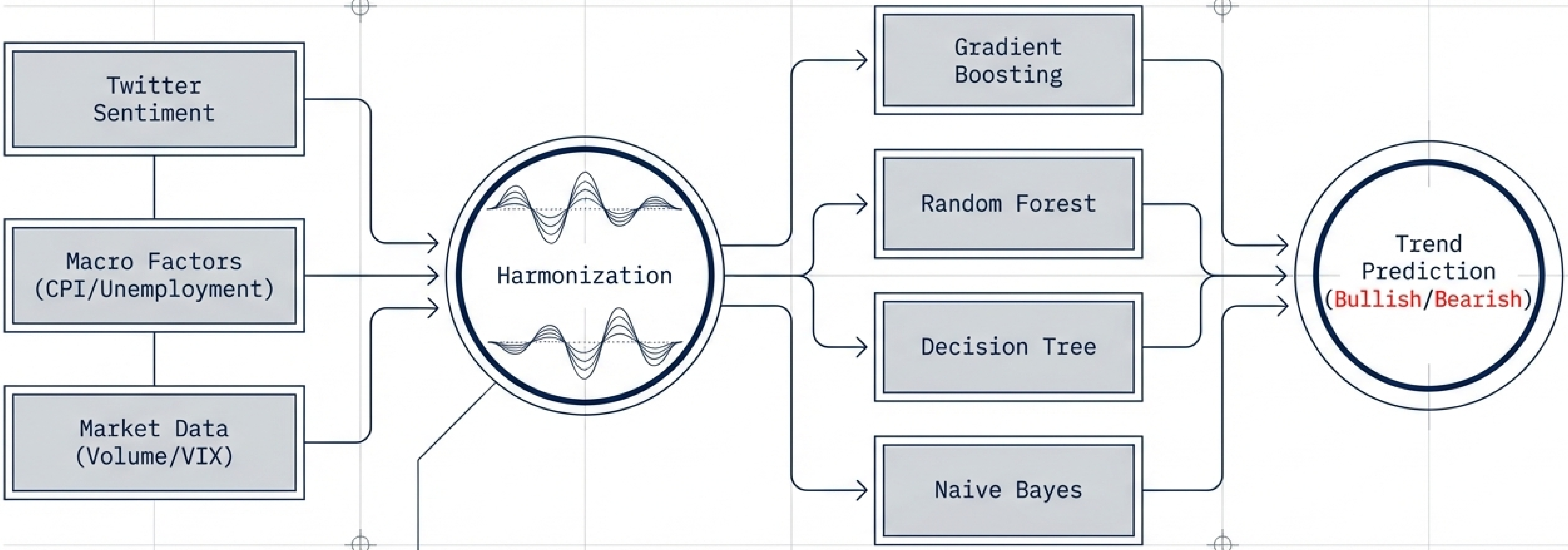


INPUT: 500,000 Tweets (Dec '22 - Mar '23)

TAGS: #ChatGPT, #AI, #GenerativeAI

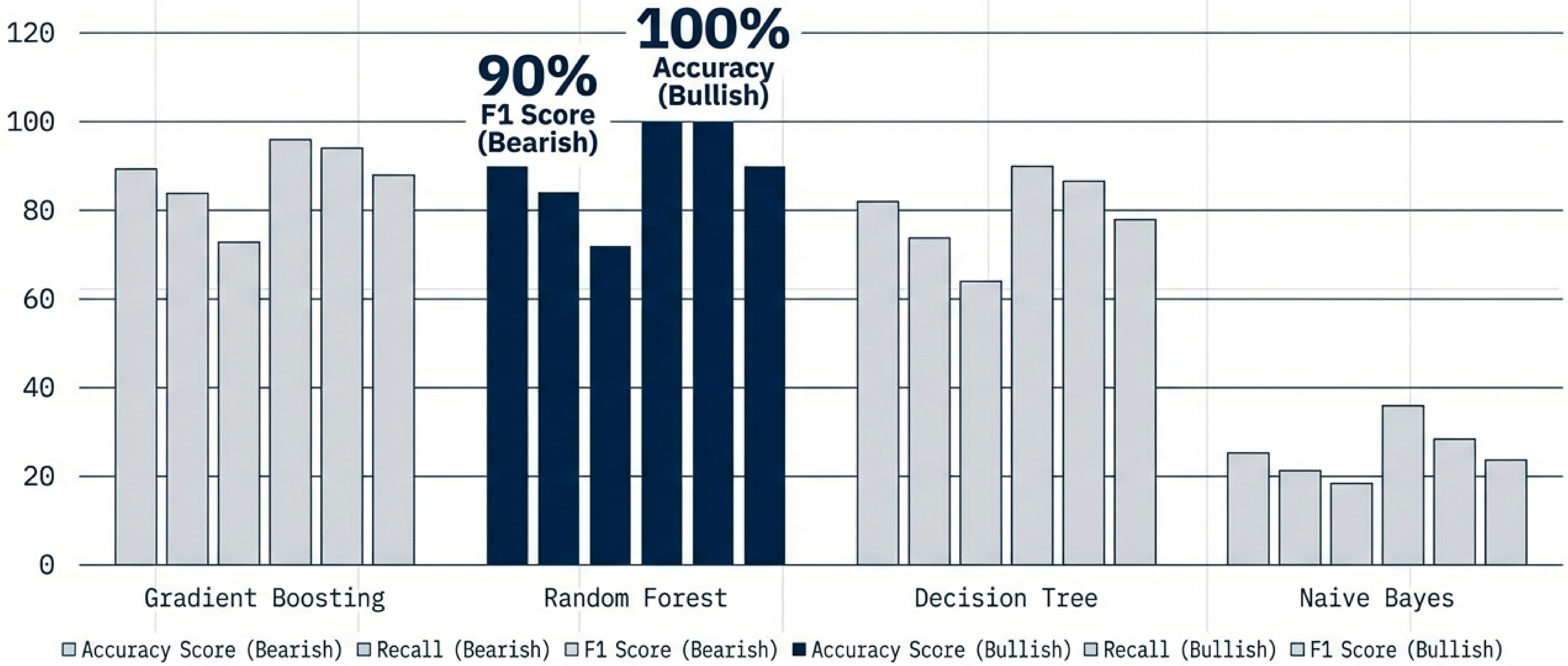
CONTROLS: CPI, Unemployment, VIX

Methodology: The Fusion Model

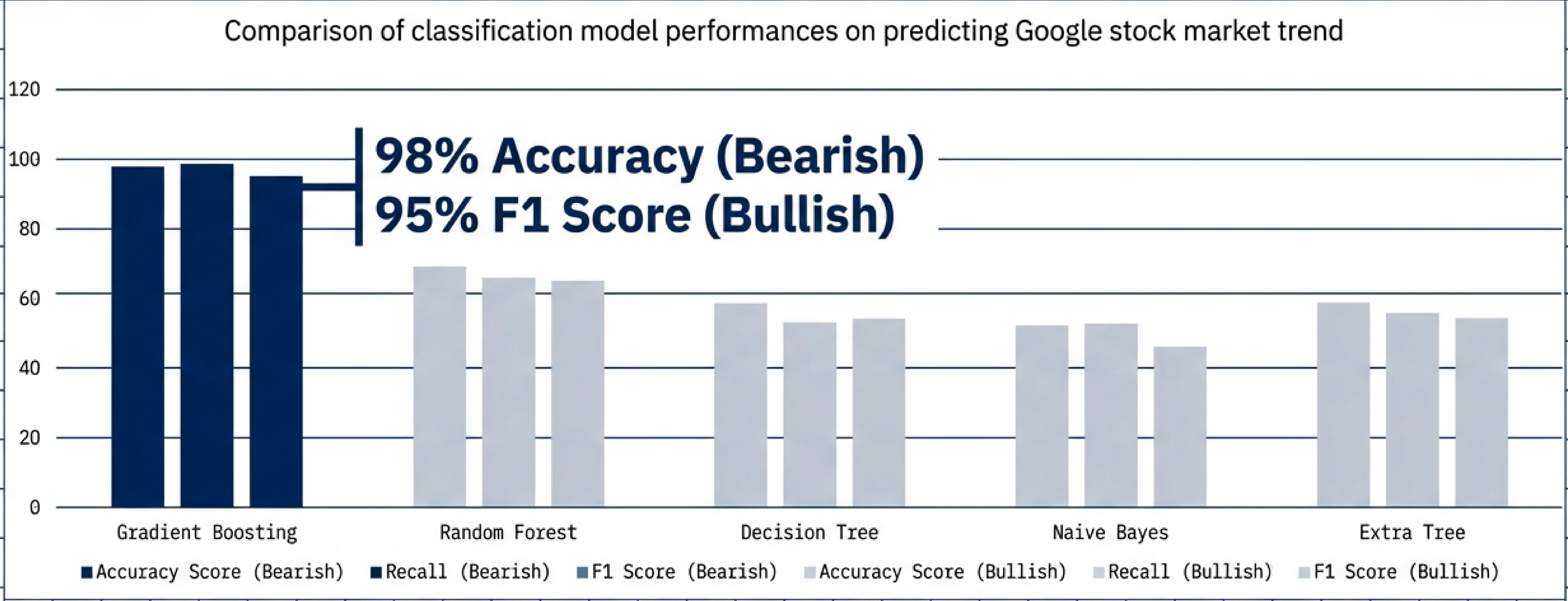


The Secret Sauce: Sentiment does not exist in a vacuum. It must be harmonized with economic reality.

The Champion: Random Forest (Microsoft)

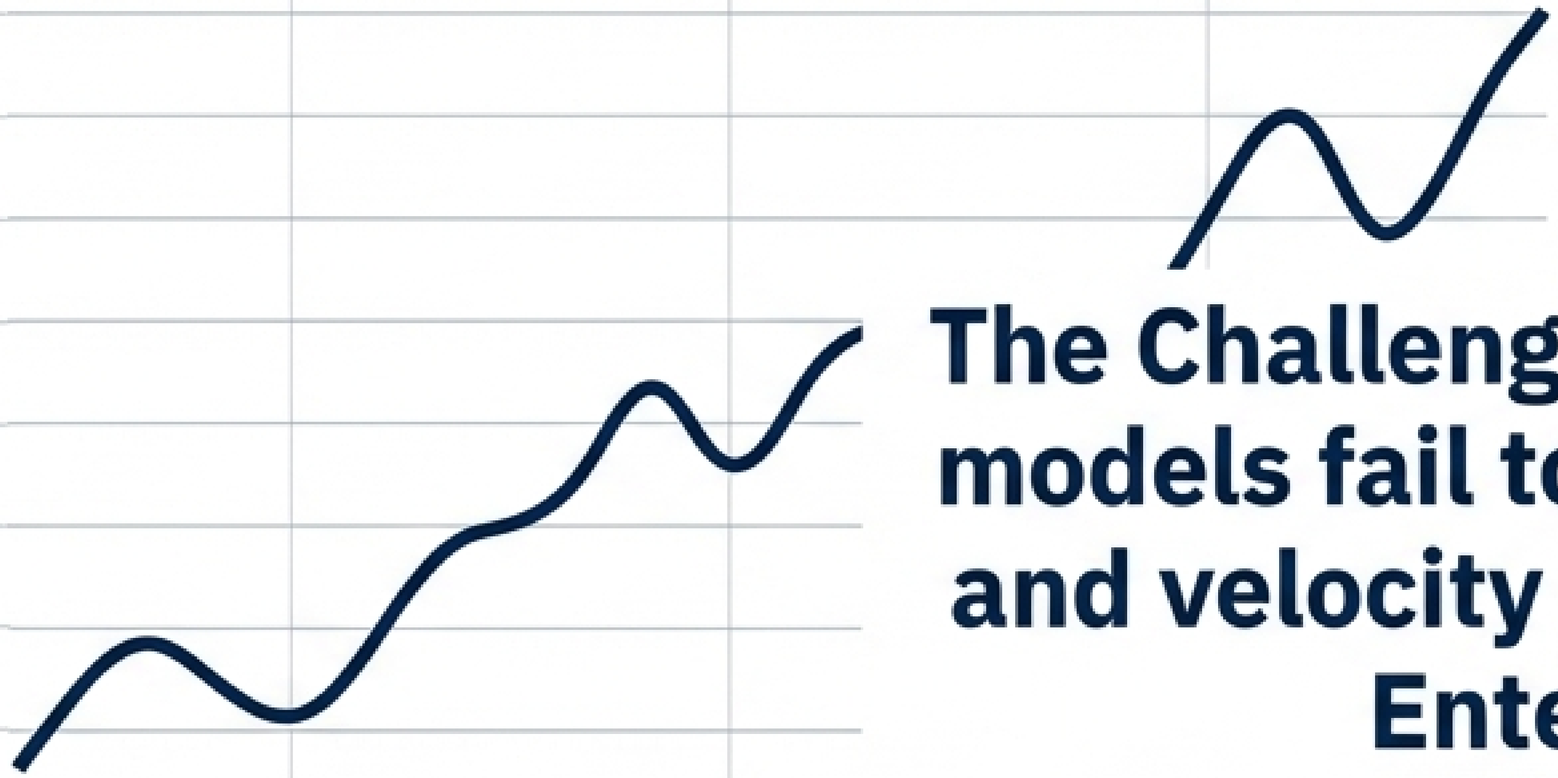


The Nuance: Gradient Boosting (Google)



Takeaway: No single algorithm rules all. Gradient Boosting captured Google's specific non-linear volatility.

Phase 2: The Cognitive Leap



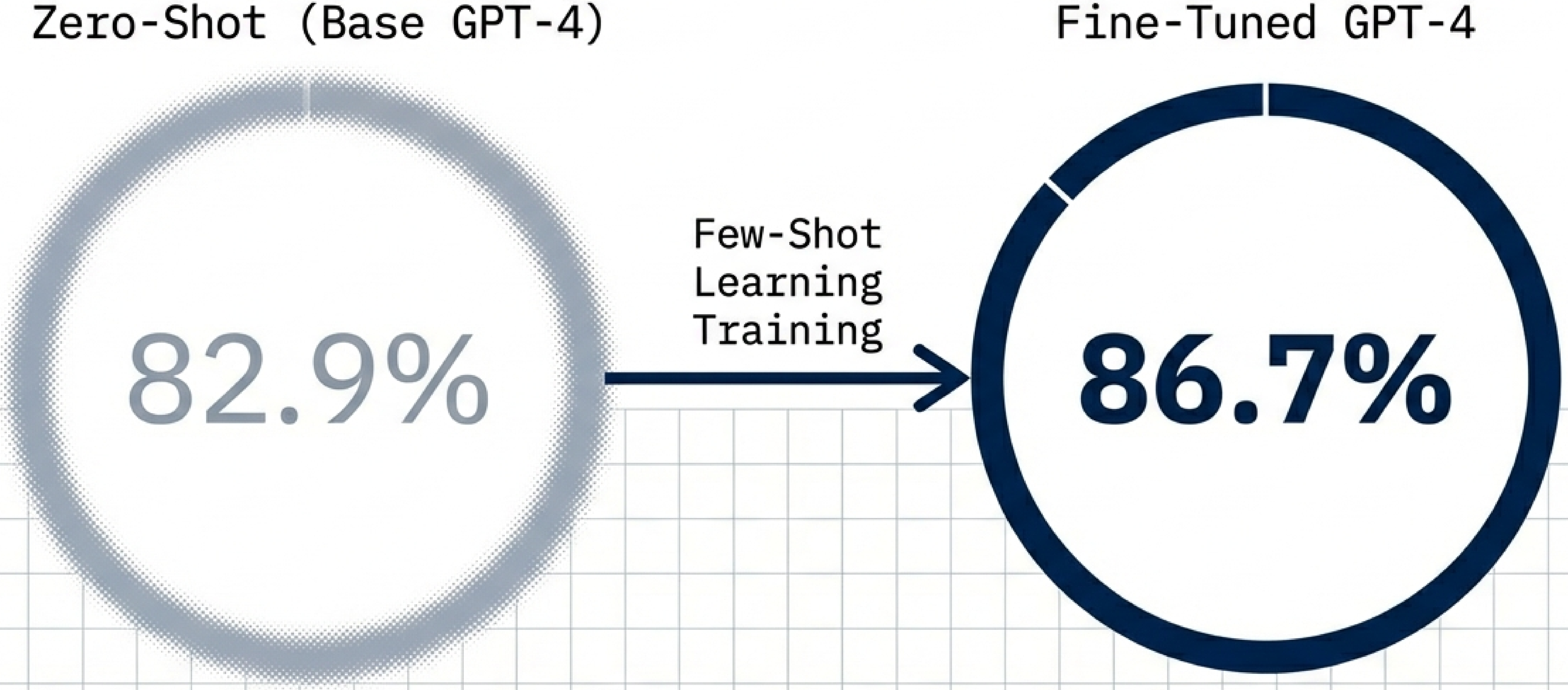
The Challenge: Traditional ML models fail to parse the slang and velocity of Crypto news. Enter LLMs.

Tech Stock
Structured Volatility

Cryptocurrency
Extreme Volatility

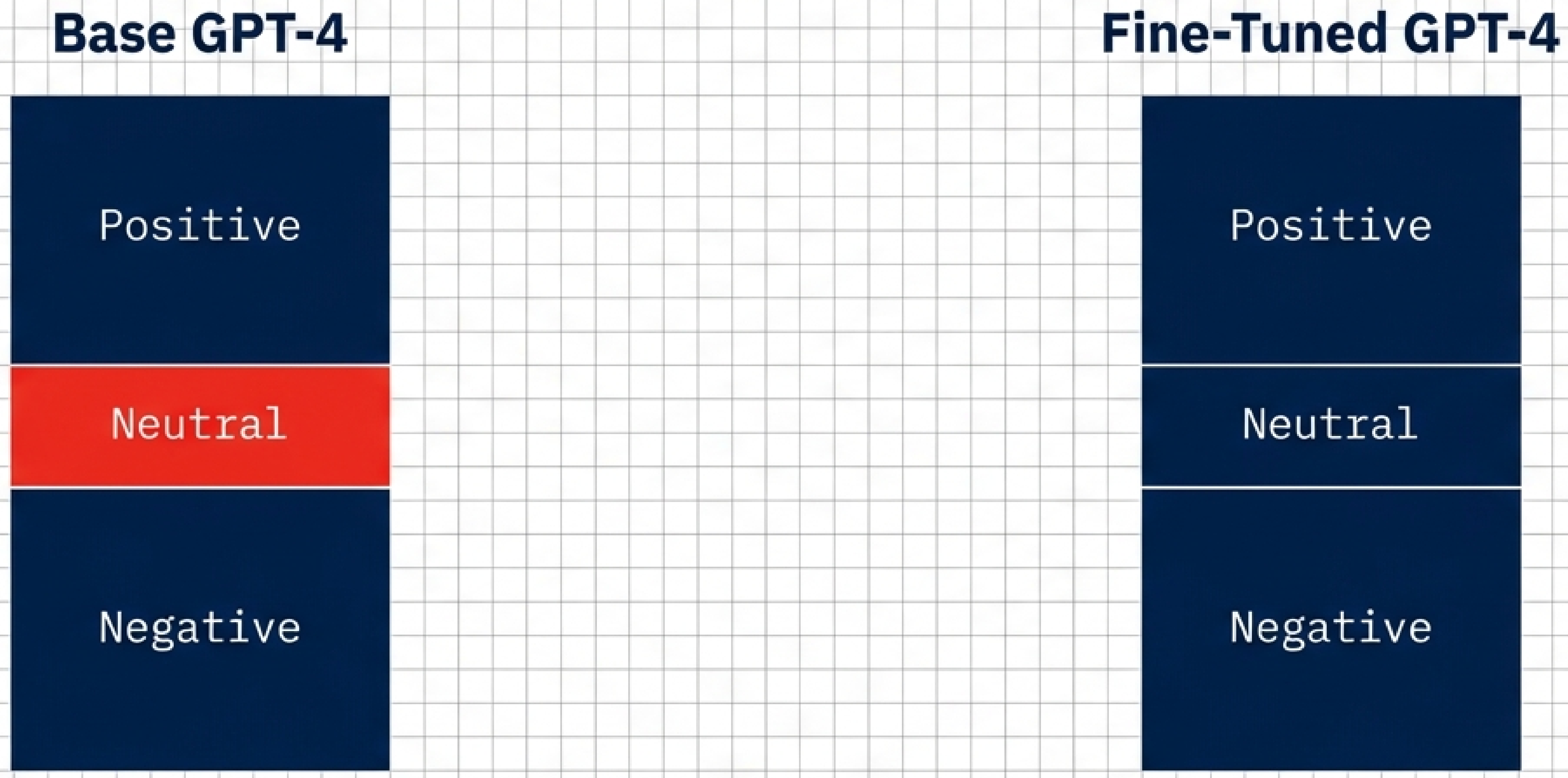
GPT-4 vs. BERT vs. FinBERT

Transforming the Generalist into a Specialist



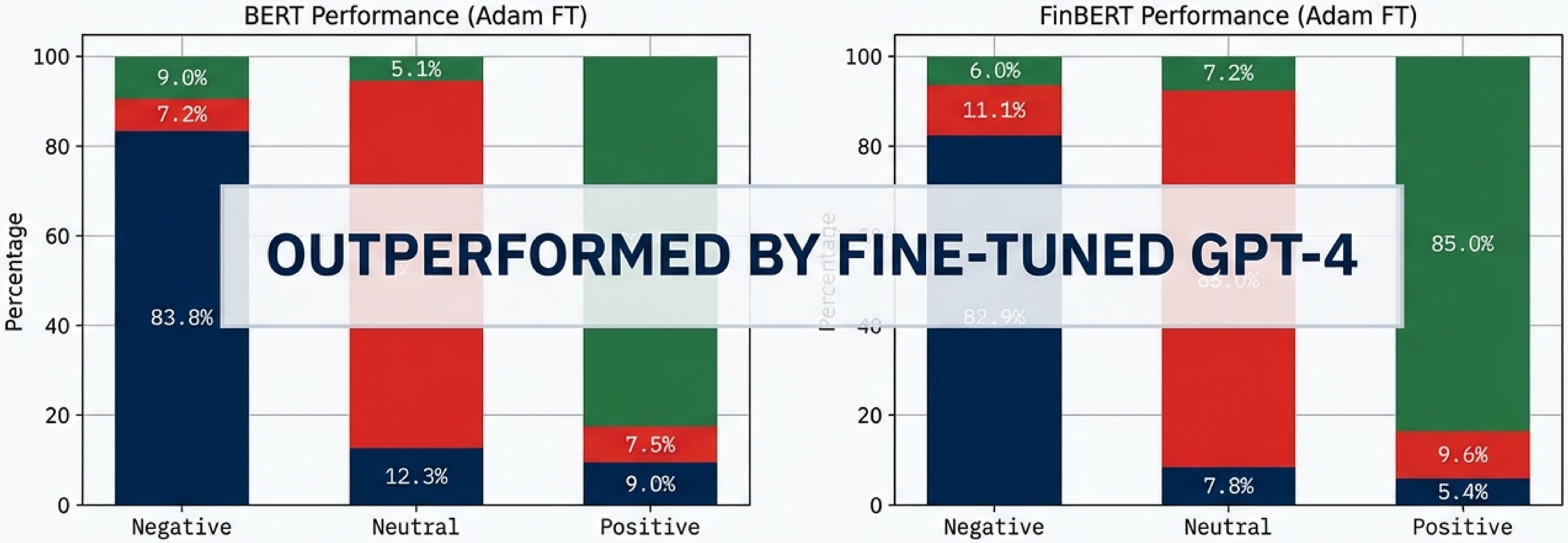
Fine-tuning outperforms even the domain-specific FinBERT (84.3%).

Mastering the Neutral Signal



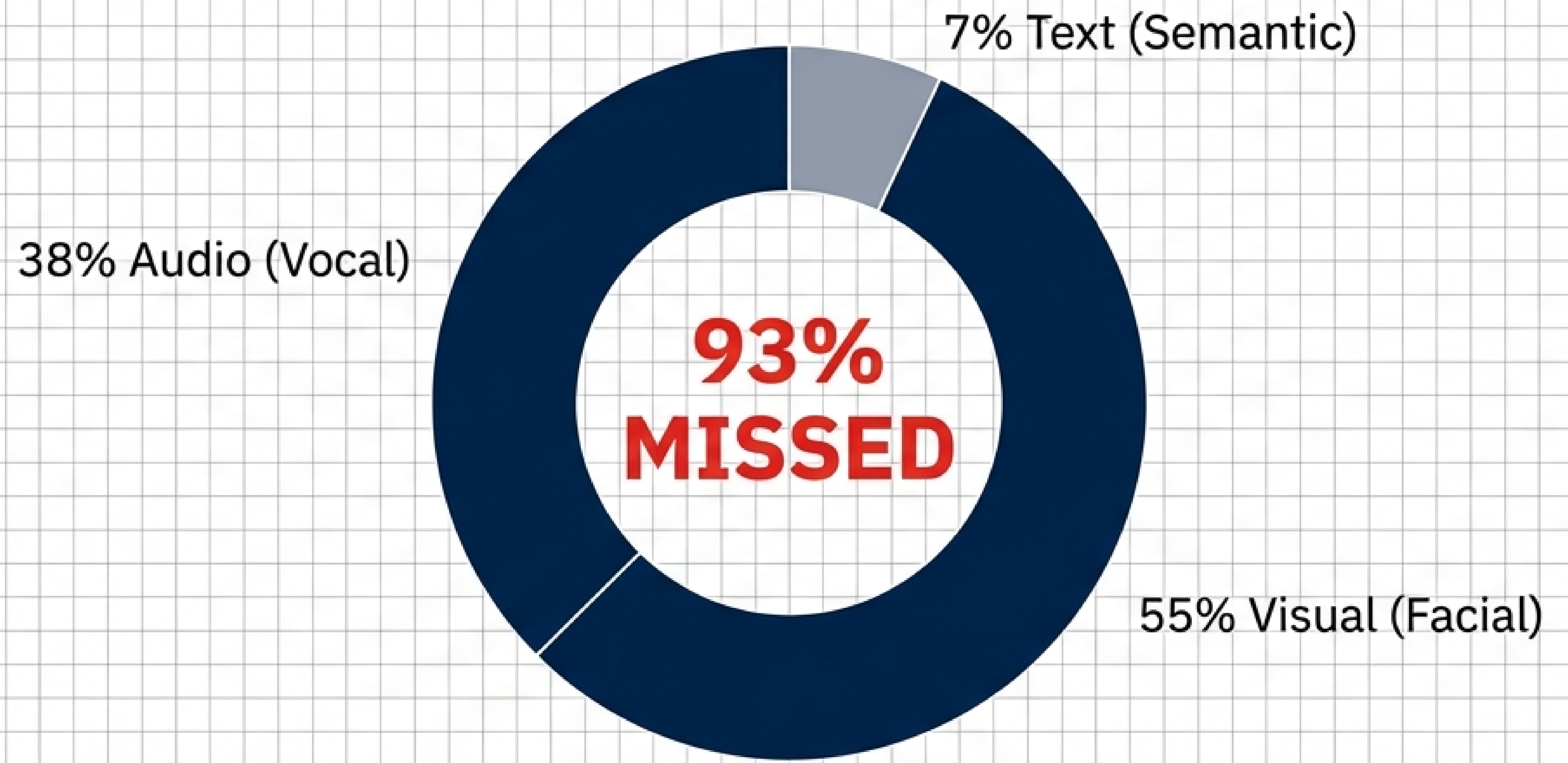
Key Insight: Base models mistake neutrality for negativity. Fine-tuning corrects the 'Neutral' signal fidelity.

GPT-4 vs. BERT: The New Standard



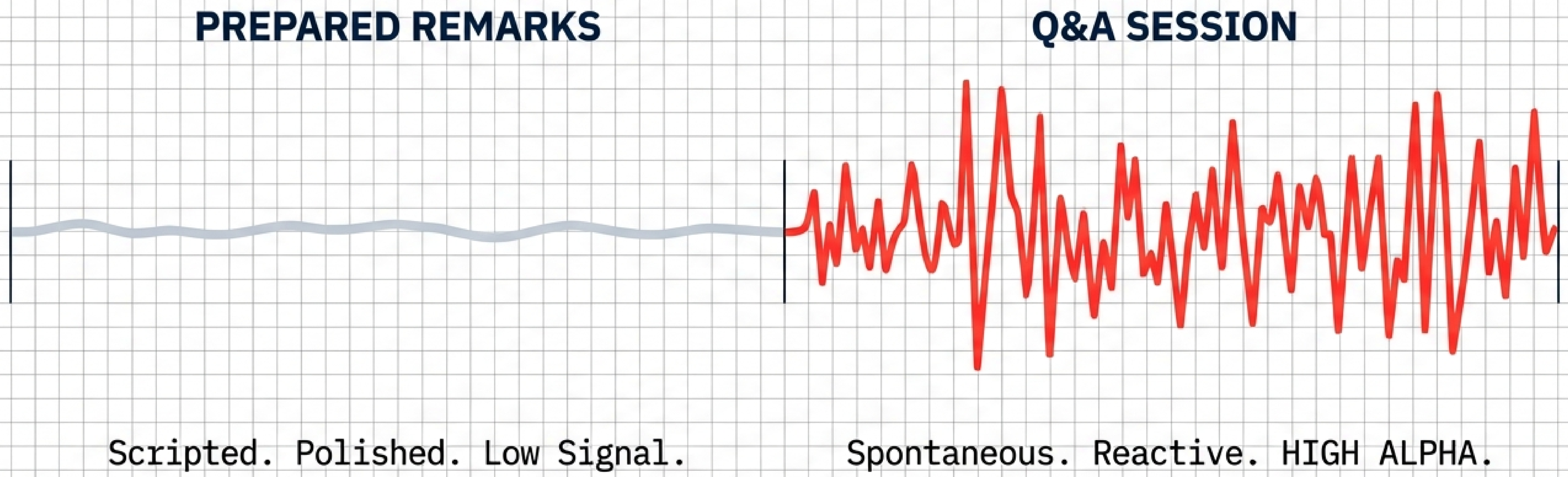
Fine-tuned LLMs provide the highest fidelity signal for crypto sentiment, surpassing specialized financial models.

The Limit of Text



The Mehrabian Rule: By analyzing only text, we miss the vast majority of the human signal.

The Earnings Call: A Tale of Two Signals



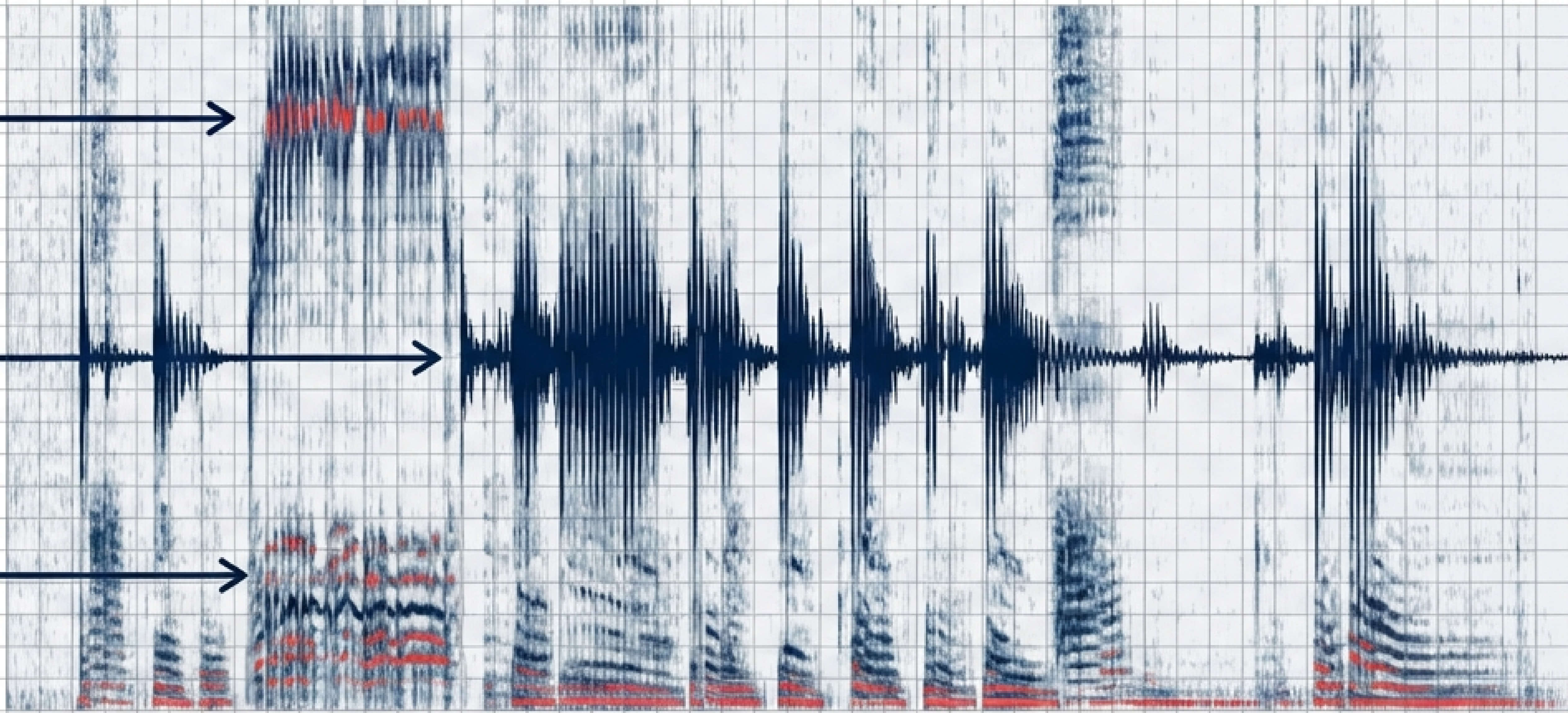
Research confirms the Q&A section holds significantly higher predictive power for abnormal returns.

Listening to the Voice (Paralinguistics)

PITCH: High pitch correlates with nervousness/low credibility.

INTENSITY: Volume fluctuations indicating stress.

JITTER & SHIMMER: Micro-tremors in frequency.



+1 SD in Positive Vocal Affect = +7.53 bps Unexpected Earnings (90 Days).

Swiss Financial Precision

Detecting Deception

OMISSION:
Silence on shareholder value.

VOCAL FLATTENING:
Fewer non-extreme positive words.

DEFLECTION:
References general "macro economy" to avoid company specifics.

Linguistic models detecting these traits outperform models based purely on accounting ratios.

Whose Voice Matters?

MANAGERS

OPTIMISM OPTIMISM OPTIMISM
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Consistently optimistic.
Less complex.
Lower market impact.

IBM Plex Mono Regular

ANALYSTS

NEUTRAL

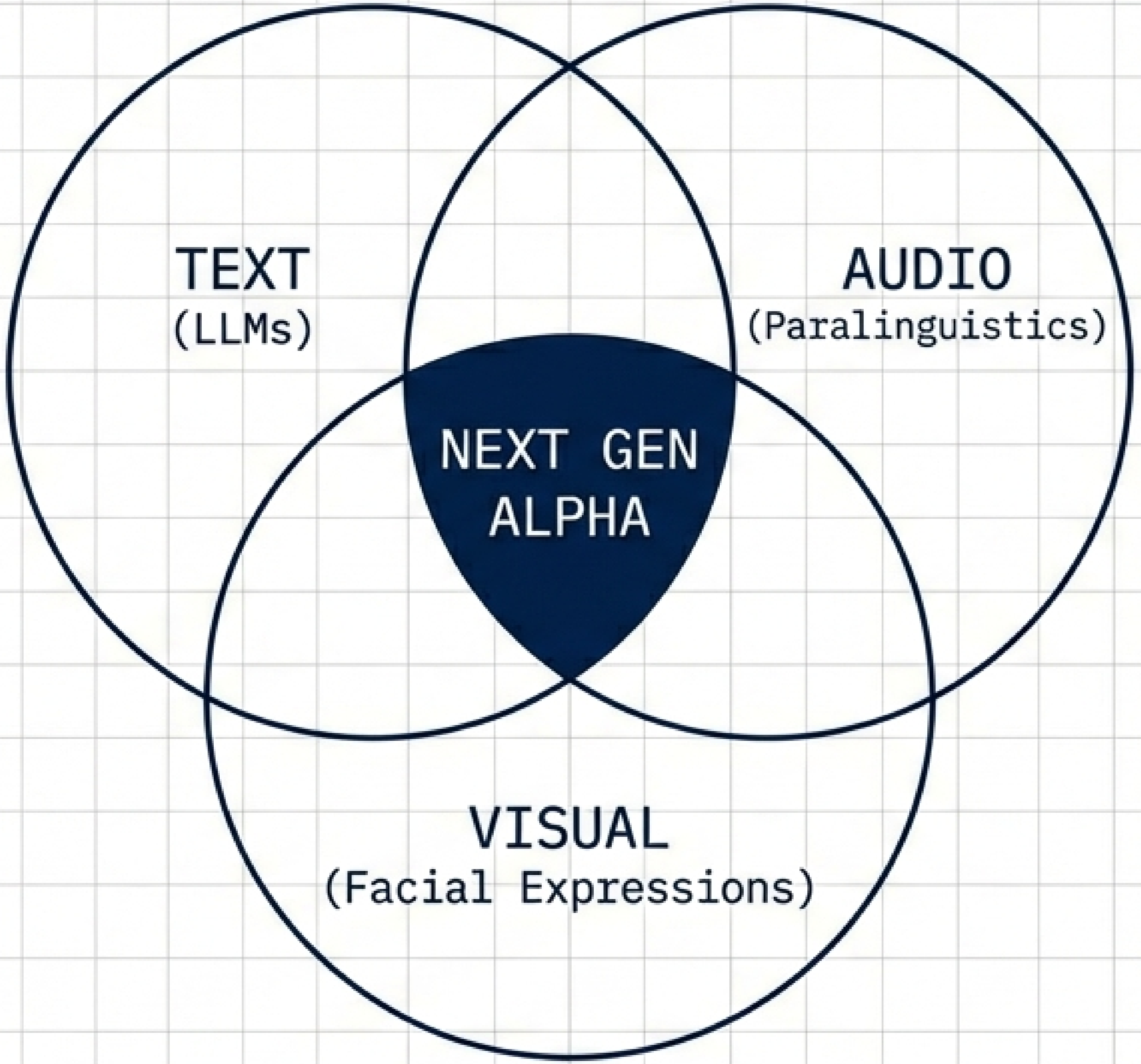
PRAISE

Objective tone.
High market impact.

Analyst 'Praise' = +1.34% Positive Abnormal Returns.

IBM Plex Mono Regular

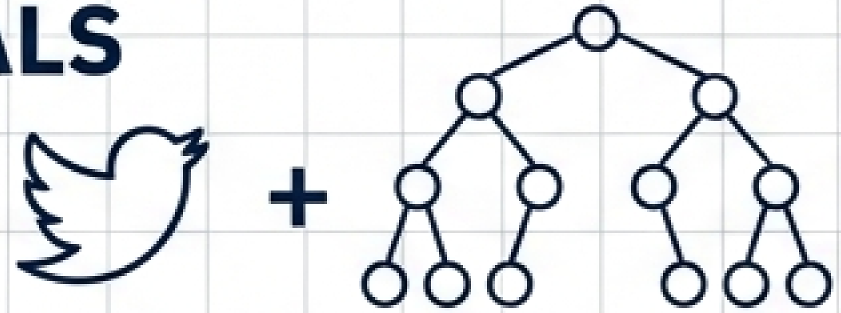
The Future is Multimodal



“All dual combinations of data outperform all singular modality models.” — Poria et al.

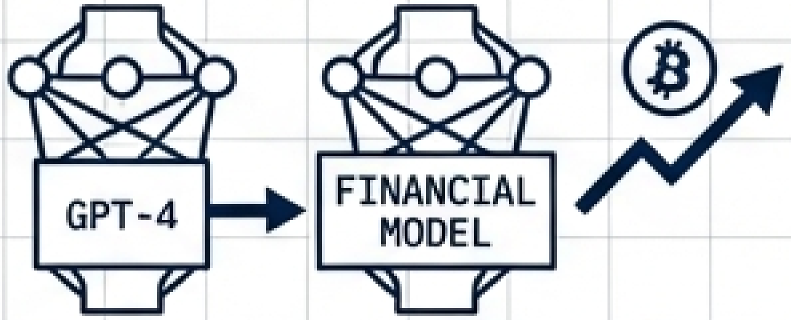
Summary of Insights

1. SOCIAL SIGNALS



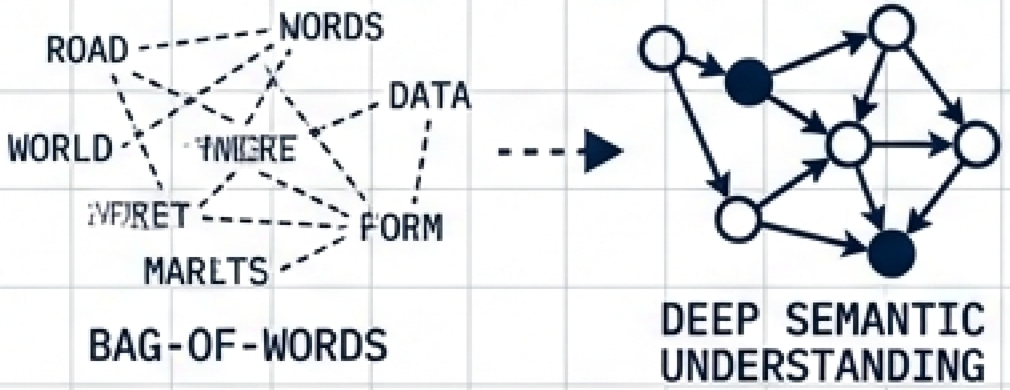
Twitter sentiment + Random Forest accurately forecasts **Big Tech trends**.

2. FINE-TUNING



GPT-4 outperforms specialized financial models in **Crypto** when fine-tuned.

3. CONTEXT



We must move beyond “bag-of-words” to **deep semantic understanding**.

4. SENSORY FUTURE

Combining text with **audio analysis** opens the next frontier of **financial truth**.

In a market of noise, the best listener wins.

References

1. Amin, M. S., et al. (2024). Harmonizing Macro-Financial Factors and Twitter Sentiment Analysis in Forecasting Stock Market Trends. Journal of Computer Science and Technology Studies.
2. Roumeliotis, K. I., et al. (2024). LLMs and NLP Models in Cryptocurrency Sentiment Analysis: A Comparative Classification Study. Big Data Cogn. Comput.
3. Todd, A., et al. (2024). Text-based sentiment analysis in finance: Synthesising the existing literature and exploring future directions. Intelligent Systems in Accounting, Finance and Management.